

Assignment on the Following Problems

1. A point charge of $+3.12 \times 10^{-6} \text{ C}$ is 12.3 cm distant from a second point charge of $-1.48 \times 10^{-6} \text{ C}$. Calculate the magnitude of the force on each charge.
2. What must be the distance between point charge $q_1 = 26.3 \mu\text{C}$ and point charge $q_2 = -47.1 \mu\text{C}$ in order that the attractive electrical force between them has a magnitude of 5.66 N?
3. An alpha particle, the nucleus of a helium atom, has a mass of $6.64 \times 10^{-27} \text{ kg}$ and a charge of $+2e$. What are the magnitude and direction of the electric field that will balance its weight?
4. In a uniform electric field near the surface of the Earth, a particle having a charge of $-2.0 \times 10^{-9} \text{ C}$ is acted on by a downward electric force of $3.0 \times 10^{-6} \text{ N}$. (a) Find the magnitude of the electric field. (b) What is the magnitude and direction of the electric force exerted on a proton placed in this field? (c) What is the gravitational force on the proton? (d) What is the ratio of the electric force to the gravitational force in this case?
5. What is the magnitude of a point charge chosen so that the electric field 75.0 cm away has the magnitude 2.30 N/C?
6. Calculate the dipole moment of an electron and a proton 4.30 nm apart.
7. Calculate the magnitude of the electric field, due to an electric dipole of dipole moment $3.56 \times 10^{-29} \text{ C} \cdot \text{m}$, at a point 25.4 nm away along the bisector axis.
8. Find the electric field at the center of the square of Fig. 21. Assume that $q = 11.8 \text{ nC}$ and $a = 5.20 \text{ cm}$.

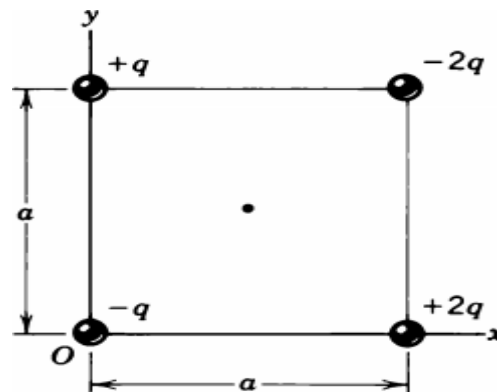


Figure 21

9. What is the magnitude of an isolated positive point charge be for the electric potential at 10 cm from the charge to be +100 volts?
10. What is the electric potential at the surface of a gold nucleus? The radius is 6.6×10^{-15} m and the atomic number $Z = 79$.
11. Two protons in a nucleus of U^{235} are 6.0×10^{-15} m apart. What is their mutual electric potential energy?
12. Three charges are arranged on in Fig. what is their mutual potential energy? Assume that $q = +1.0 \times 10^{-7}C$ & $a = 10$ cm.
13. What is the potential at the center of the square in figure? Assume that $q_1 = +1.0 \times 10^{-8}C$; $q_2 = -2.0 \times 10^{-8}C$; $q_3 = +3.0 \times 10^{-8}C$ and $q_4 = +2.0 \times 10^{-8}C$ & $a = 1$ meter.
14. A parallel plate capacitor is charged to $50 \mu C$ at 150 V. It is then connected to another capacitor of capacitance 4 times the capacitance of the first capacitor. Find the loss of energy.
15. An air capacitor has an area of 100 cm^2 , the plates being 4mm apart. It is connected in series with another capacitor having a plate area of 20 cm^2 , the plates being separated by a dielectric 0.1 mm thick having a relative permittivity of 2.5. If the potential difference across the air capacitor is 100 V Find the voltage across the series combination.
16. Three capacitor A, B, C has capacitance 10,50 and $25 \mu F$ respectively. Calculate (i) charge on each when connected in parallel to a 150 V supply (ii) total capacitance and (iii) potential difference across each when connected in series.
17. Two capacitor of capacitance $4 \mu F$ and $2 \mu F$ respectively are joined in series with a battery of e.m.f 100V. The connection are broken and the like terminals of the capacitor are then joined. Find the final charge on each capacitor.
18. The parallel plate of an air filled capacitor are everywhere 1.0 mm apart. What must the plate area lie if the capacitance is to be 1.0 farad ?
19. A capacitor C_1 is charged to a potential difference V_0 . this charging battery is then removed and the capacitor is connected with another uncharged C_2 .(a) What is the final potential difference V across the combination.(b) What is the stored energy before and after connecting the second capacitor.
20. Two capacitor of capacitance $2.12 \mu F$ and $3.88 \mu F$ respectively are joined in series with a battery of e.m.f 328V. Calculate the total energy stored in capacitor.